MNRAS 000, 1–12 (2026) Preprint 14 October 2026 Compiled using MNRAS LATEX style file v3.0

Season by season SURD analysis of the continuum and velocity-resolved Hβ emission of NGC 5548

Ayan Mitra,[1]★ Vasilios Zarikas[2]
[1]Department of Astronomy, University of Illinois Urbana-Champaign, Urbana, IL 61801, USA
[2]Department of Physics, University of Thessaly, Lamia 35100, Greece

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We apply the Synergistic–Unique–Redundant Decomposition (SURD) to the five historical monitoring seasons (1989–1993) of the Seyfert 1 galaxy NGC 5548, following the season by season strategy of recent Lijiang campaigns. For each season we take the 5100 Å continuum and the total, blue, core and red Hβ light curves in two four-variable sets, use each variable in turn as the target, and decompose its future into 26 unique, redundant and synergistic components and an information leak, as functions of lag from 1 to 30 d, for common and for uncommon predictor lags. The leak is lowest at the shortest lag (36 of 40 curves), where the history of the target dominates, and highest beyond 20 d. The unique continuum contribution to the line targets peaks at 16–29 d in 13 of 15 curves, a median 6 d longer than the published Hβ lags of 11.0–19.7 d. The red wing peaks before the blue wing in four of the five seasons, the red-leads-blue ordering that has been associated with inflow in 2019 and 2021, and the order is reversed in 1990. The allocation among components depends strongly on the number of histogram bins (median total variation 0.27–0.58) and the mean leak follows the number of samples, so the results are descriptive; no significance test or flux-error propagation is applied.

Key words: galaxies: active – galaxies: nuclei – galaxies: individual: NGC 5548 – methods: statistical

1 INTRODUCTION

Reverberation mapping uses the delayed response of the broad emission lines to continuum variability to measure the size and kinematics of the broad-line region (BLR) (Peterson et al. 2004). NGC 5548 is the best monitored object: it has been observed since 1988 and its BLR changes from season to season. Lu et al. (2022) and Xi et al. (2025) divide the Lijiang monitoring into seasons and show that the velocity-resolved lag profile of Hβ evolves from a symmetric 'M' shape in 2015, through a red-leads-blue pattern in 2019 and 2021 interpreted as inflow, to a blue-leads-red pattern in 2020 and 2023 interpreted as outflow. These conclusions rest on cross-correlation lags between the continuum and equal flux velocity bins of the line.

SURD (Martínez-Sánchez et al. 2024) asks a different question. It splits the information that a set of observed variables carries about the future of a target into unique, redundant and synergistic parts, and reports the fraction that the observed variables do not explain as a leak. Applied to the continuum and the blue, core and red parts of Hβ, it can show which of them carries information on the future of another, and at which lag. The Lijiang spectra of the modern seasons are not public. We therefore apply the season by season strategy of Lu et al. (2022) and Xi et al. (2025) to the five historical seasons of the AGN Watch campaign, for which the corrected line fluxes are published (Wanders & Peterson 1996).

Section 2 summarizes SURD. Section 3 describes the data, the analysis and the results, and compares them with the published lags and velocity-resolved profiles. Section 4 gives the conclusions.

2 SYNERGISTIC–UNIQUE–REDUNDANT DECOMPOSITION (SURD) FOR TIME SERIES

2.1 Problem formulation

Consider a collection of N observed stochastic processes,

$$\mathbf{Q}(t) = \big(Q_1(t), Q_2(t), \ldots, Q_N(t)\big), \tag{1}$$

sampled at discrete times $t = t_n$. For a chosen target index j, the goal is to quantify how much information about the future state

$$Q_j^+ \equiv Q_j(t + \Delta T) \tag{2}$$

is contained in the observed variables at time t and, crucially, to decompose that predictive information into *redundant*, *unique*, and *synergistic* contributions (Martínez-Sánchez et al. 2024).

The central motivation is that in multivariate systems, pairwise predictive measures such as Granger causality or transfer entropy can establish that a variable is informative about a target, but they do not in general distinguish whether that information is:

(i) unique to one predictor,
(ii) duplicated across several predictors, or
(iii) available only from a joint observation of several predictors.

SURD is designed precisely to perform that decomposition in a time-series setting (Martínez-Sánchez et al. 2024; Schreiber 2000; Williams & Beer 2010).

★ E-mail: ayan@illinois.edu

© 2026 The Authors

2.2 Information-theoretic preliminaries

Let $Y \equiv Q_j^+$ denote the target and let $\mathbf{X}$ denote the vector of observed predictors. For discrete variables, the Shannon entropy is

$$H(Y) = -\sum_y p(y)\log_2 p(y), \tag{3}$$

and the conditional entropy of Y given $\mathbf{X}$ is

$$H(Y \mid \mathbf{X}) = -\sum_{y,\mathbf{x}} p(y,\mathbf{x})\log_2 p(y \mid \mathbf{x}). \tag{4}$$

The mutual information between Y and $\mathbf{X}$ is

$$I(Y;\mathbf{X}) = H(Y) - H(Y \mid \mathbf{X}) = \sum_{y,\mathbf{x}} p(y,\mathbf{x})\log_2 \frac{p(y,\mathbf{x})}{p(y)p(\mathbf{x})}. \tag{5}$$

This quantity measures the total predictive information about the future target that is available in the observed present state (Shannon 1948; Cover & Thomas 2006).

In SURD, the unexplained part of the future is interpreted as an *information leak*,

$$\Delta I_{\text{leak}\to j} = H(Y \mid \mathbf{X}), \tag{6}$$

so that the forward information balance reads

$$H(Y) = I(Y;\mathbf{X}) + \Delta I_{\text{leak}\to j}. \tag{7}$$

Equation (7) is one of the core ideas of SURD: the future uncertainty of the target splits into an observed part, explained by the measured variables, and an unobserved part, attributed to hidden or unmeasured influences (Martínez-Sánchez et al. 2024).

2.3 From total predictive information to information-decomposition components

Let the predictor vector be

$$\mathbf{X} = (X_1, \ldots, X_M), \tag{8}$$

where the X_α may correspond either to different physical variables at the same time, or to different lagged copies of the same variables. SURD decomposes the observable predictive information $I(Y;\mathbf{X})$ into additive components:

$$I(Y;\mathbf{X}) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j}, \tag{9}$$

where:

- $\Delta I^R_{\mathcal{A}\to j}$ is the *redundant* contribution from the group of predictors indexed by $\mathcal{A}$,
- $\Delta I^U_{\alpha\to j}$ is the *unique* contribution of predictor X_α,
- $\Delta I^S_{\mathcal{A}\to j}$ is the *synergistic* contribution from the group $\mathcal{A}$,
- C denotes all non-singleton subsets of $\{1,\ldots,M\}$.

The full future entropy therefore obeys

$$H(Y) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j} + \Delta I_{\text{leak}\to j}. \tag{10}$$

The three information-decomposition types have the following interpretation (Martínez-Sánchez et al. 2024):

- Redundant: the same predictive information is shared across several predictors,
- Unique: a predictor contains information not available from any other predictor alone,
- Synergistic: information appears only when several predictors are observed jointly.

For the three-predictor scans below, "total synergy" denotes $S_{12}+S_{13}+S_{23}+S_{123}$ and "total redundancy" denotes $R_{12}+R_{13}+R_{23}+R_{123}$. These sums, rather than any single atom, are the quantities shown in the primary lag scans.

The statewise construction and detailed lagged formalism follow Martínez-Sánchez et al. (2024); we omit that derivation here and retain only the quantities required for the analysis.

2.4 Normalization and interpretation

Because the observable predictive information is conserved by construction, the unique, redundant, and synergistic terms are naturally normalized by the total observable mutual information:

$$\begin{aligned}
\widehat{\Delta I}^R_{\mathcal{A}\to j} &= \frac{\Delta I^R_{\mathcal{A}\to j}}{I(Y;\mathbf{X})},\\
\widehat{\Delta I}^U_{\alpha\to j} &= \frac{\Delta I^U_{\alpha\to j}}{I(Y;\mathbf{X})},\\
\widehat{\Delta I}^S_{\mathcal{A}\to j} &= \frac{\Delta I^S_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}.
\end{aligned} \tag{11}$$

These normalized terms sum to unity:

$$\sum_{\mathcal{A}\in C} \widehat{\Delta I}^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \widehat{\Delta I}^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \widehat{\Delta I}^S_{\mathcal{A}\to j} = 1. \tag{12}$$

The leak is normalized instead by the total entropy of the target:

$$\widehat{\Delta I}_{\text{leak}\to j} = \frac{\Delta I_{\text{leak}\to j}}{H(Y)} = \frac{H(Y \mid \mathbf{X})}{H(Y)}. \tag{13}$$

Hence:

- $\widehat{\Delta I}_{\text{leak}\to j} \approx 0$ means the observed predictors account for almost all predictive information about the target,
- $\widehat{\Delta I}_{\text{leak}\to j} \approx 1$ means most of the target's future uncertainty remains unexplained by the observed variables.

2.5 Practical estimation

In applications, the probabilities entering Eq. (5) and the statewise SURD construction must be estimated from finite data. The original SURD work uses histogram-based and transport-map-based estimators depending on dimensionality and sample size (Martínez-Sánchez et al. 2024). In a histogram implementation, one discretizes the predictor-target space into bins and estimates

$$p(y,\mathbf{x}) \approx \frac{n(y,\mathbf{x})}{N_{\text{samples}}}, \tag{14}$$

where $n(y,\mathbf{x})$ is the bin count. The resulting empirical joint distribution is then used to compute all required marginal and conditional probabilities. More sophisticated estimators, such as transport maps, are advantageous in high dimensions because the number of bins required by direct histograms grows exponentially with the number of predictors.

Algorithmically, the SURD workflow for a target Q_j^+ is:

(i) choose the predictor vector $\mathbf{X}$, including variables, lags, or contemporaneous channels;
(ii) estimate the probability distributions $p(y, \mathbf{x}_{\mathcal{A}})$ for all relevant predictor subsets $\mathcal{A}$;
(iii) compute the specific informations $\tilde{\imath}_{\mathcal{A}}(y)$ for every target state y;
(iv) decompose the statewise increments into redundant, unique, and synergistic parts;
(v) average over y to obtain ΔI^R, ΔI^U, and ΔI^S;
(vi) compute the leak from Eq. (6);
(vii) normalize the observable components by $I(Y;\mathbf{X})$ and the leak by $H(Y)$.

3 SURD COMPONENTS IN THE FIVE HISTORICAL NGC 5548 SEASONS

3.1 Series, epochs and analysis design

We apply the seasonal strategy of Lu et al. (2022) and Xi et al. (2025) to the historical monitoring of NGC 5548 between 1988 and 1993. Their seasons are analysed independently; we do the same for the five historical seasons, which carry the labels Year 1 to Year 5 in Table 6 of Lu et al. (2022). The epochs are the intervals of Table 2 of Peterson et al. (2004): JD less 2400000 from 47509 through 47809 (1989), 47861 through 48179 (1990), 48225 through 48534 (1991), 48623 through 48898 (1992) and 48954 through 49255 (1993).

Five observed series are prepared for each epoch: the 5100 Å continuum (F5100), the total broad Hβ flux, and its blue, core and red components. The line measurements are the corrected published columns of Wanders & Peterson (1996), with blue, core and red defined over the observed wavelength intervals 4870–4920, 4920–4960 and 4960–5010 Å. These fixed wavelength intervals play the role of the equal flux velocity bins that Xi et al. (2025) construct over 4700–5050 Å; they are not equal flux bins. Observations on the same date are combined by their mean. The continuum retains its host galaxy contribution. Total and component fluxes come from the same spectra, and the total is the sum of its components, so they are not four independent measurements.

Two variable sets are analysed, each with four members: total, blue, core and red Hβ; and F5100, blue, core and red Hβ. Each member becomes the future target Y in turn. All four members, including the past of the target itself, are the predictors $\mathbf{X}$. The unchanged SURD implementation of Martínez-Sánchez et al. (2024) is used throughout. For four predictors the decomposition has four unique components U, eleven redundant components R (six pairs, four triples and one set of four predictors) and eleven synergistic components S with the same combinations. These are 26 components rather than 25. Following Figure 4.1 of Martínez Sánchez (2025), each component is divided by the joint mutual information between the future target and the whole predictor vector,

$$f_a(\tau) = \frac{I_a(Y;\mathbf{X}_\tau)}{I(Y;\mathbf{X}_\tau)}, \qquad \sum_{a=1}^{26} f_a(\tau) = 1, \tag{15}$$

where I_a is the contribution of component a averaged over the states of the target (equations 3.16 of the thesis, ordered as in equation 3.17). The causality leak of equation 3.6 is normalized by the entropy of the target,

$$\ell(\tau) = \frac{H(Y \mid \mathbf{X}_\tau)}{H(Y)}, \tag{16}$$

and shown in its own panel. The component sums equal one to within 1.6×10^{-15} in all 7200 decompositions.

3.1.0.1 Regular grid and lags.

Every series is linearly interpolated onto a daily grid that spans its epoch. Interpolation never extrapolates and never bridges a gap between neighbouring observations longer than 15 days. The target is $Y(t)$ and the predictors are $X_i(t - L - \delta_i)$, with base lag L from 1 through 30 days in one day steps. The 30 day span covers the common lag ranges proposed for the modern seasons. Probabilities are histograms with two bins and, as a sensitivity setting, three bins. Bin boundaries are the marginal quantiles of each full native epoch series. Table 1 lists the data support. Every lag of every curve retains at least 118 tuples. Grid tuples are interpolated and autocorrelated, so they are not independent samples.

3.1.0.2 Uncommon predictor lags (equation 3.18).

Equation 3.18 of the thesis builds the predictor vector from copies of each variable at different past times. We give every predictor variable its own lag $L + \delta_i$, so that the four predictors still define the 26 components. The common lag analysis has $\delta_i = 0$. In the *blue leading* configuration the blue wing is sampled $\delta = 5$ days further back and in the *red leading* configuration the red wing is. The component that responds earlier to a continuum event carries that event further in its past, so this is the arrangement that aligns the wings under the two kinematic patterns that Xi et al. (2025) describe. The five day offset is a stipulated analysis choice of the size of the uncertainties of published Hβ lags (two to three days); it is not a measured delay. Every configuration and lag is analysed for both variable sets, all four targets and all five epochs: 7200 decompositions, 120 figures.

3.2 Common lags: leakage and dominant components

Figures 1, 2 and 3 show three examples of the 26 normalized components and the leakage against lag. The 120 curves of every epoch, variable set, target and lag configuration are in the three atlas files `atlas_common.pdf`, `atlas_blue_leading.pdf` and `atlas_red_leading.pdf` in `collaborator_rebuild/epoch_results/figures` of the repository, one page per curve. Tables 2 and 3 give the extrema of all 40 common lag curves (two bins).

Leakage. In 36 of 40 curves the leakage minimum is at the shortest lag of one day, where the history of the target itself predicts its own future. The exceptions are the blue and core targets of the F5100 set in 1989 (minima at 8 and 24 days, leakage 0.12 and 0.05), the total target of the total set in 1990 (3 days) and the core target of the F5100 set in 1991, whose leakage reaches zero to numerical precision at 4 days. The leakage maximum lies at 20 days or longer in 35 of 40 curves. Median minimum and maximum leakage are 0.12 and 0.70 for the F5100 set and 0.09 and 0.89 for the total set. The leakage at the largest lags of the total set approaches the 0.86 to 0.98 that independent categorical variables produce in the calibration below for 90 to 512 tuples. Little predictive information remains at such lags relative to the bias level of the estimator.

Largest components. At the leakage minimum the largest component is the four-way redundancy $R(H, B, C, R)$ in 16 of the 20 curves of the total set: the components share the same spectra and sum to the total, so their past states carry the same information about the future. In the F5100 set it is a unique component (11 curves, mostly $U(F)$ and the target's own history) or a redundancy among the line components (9 curves). At the leakage maximum the largest component is a synergy in 11 of 20 total set curves and a unique component in five; in the F5100 set it is $U(F)$ in 10 curves and a synergy in seven. The largest component then carries on average 0.32 (total set) and 0.36 (F5100 set) of the joint information, against 0.55 and 0.57 at the leakage minimum.

Table 1. Data support per epoch. N_F and $N_{\rm line}$ are the numbers of native continuum and spectroscopic dates; the line number is the same for the total, blue, core and red series. The last column gives the range over lags and targets of the daily grid tuples with all five series available (common lags, two bins).

Epoch	JD−2400000	N_F	$N_{\rm line}$	Grid tuples
1989	47509–47809	124	48	134–196
1990	47861–48179	94	38	161–232
1991	48225–48534	65	32	133–211
1992	48623–48898	83	46	211–244
1993	48954–49255	191	60	231–261

Table 2. Variable set total, blue, core and red Hβ with common lags and two bins. Leakage minimum and maximum, with the lag in days in parentheses, followed by the largest unique, redundant and synergistic component over lags 1 to 30 d, with its normalized value and lag. H, B, C and R denote the total, blue, core and red series. The maxima are descriptive and have no null calibration or lag uncertainty.

Epoch	Target	$\ell_{\rm min}$ (d)	$\ell_{\rm max}$ (d)	Largest U	Largest R	Largest S
1989	Total	0.07 (1)	0.85 (25)	$U(H)$ 0.26 (1)	$R(H,B,C,R)$ 0.64 (7)	$S(B,R)$ 0.45 (27)
1989	Blue	0.16 (1)	0.85 (29)	$U(B)$ 0.36 (1)	$R(H,B,C,R)$ 0.54 (7)	$S(B,C)$ 0.38 (27)
1989	Core	0.07 (1)	0.83 (29)	$U(C)$ 0.27 (1)	$R(H,B,C,R)$ 0.62 (7)	$S(B,R)$ 0.38 (30)
1989	Red	0.17 (1)	0.88 (23)	$U(R)$ 0.22 (1)	$R(H,B,C,R)$ 0.59 (11)	$S(B,C,R)$ 0.28 (23)
1990	Total	0.04 (3)	0.91 (30)	$U(H)$ 0.16 (1)	$R(H,B,C,R)$ 0.66 (1)	$S(B,C,R)$ 0.34 (28)
1990	Blue	0.07 (1)	0.92 (30)	$U(B)$ 0.20 (1)	$R(H,B,C,R)$ 0.67 (5)	$S(B,C,R)$ 0.36 (28)
1990	Core	0.09 (1)	0.92 (28)	$U(C)$ 0.21 (1)	$R(H,B,C,R)$ 0.58 (3)	$S(B,C,R)$ 0.33 (29)
1990	Red	0.09 (1)	0.93 (28)	$U(C)$ 0.17 (15)	$R(H,B,C,R)$ 0.59 (7)	$S(C,R)$ 0.37 (28)
1991	Total	0.06 (1)	0.77 (28)	$U(C)$ 0.25 (28)	$R(H,B,C,R)$ 0.71 (3)	$S(C,R)$ 0.13 (28)
1991	Blue	0.04 (1)	0.74 (29)	$U(C)$ 0.25 (28)	$R(H,B,C,R)$ 0.70 (4)	$S(C,R)$ 0.13 (28)
1991	Core	0.02 (1)	0.68 (30)	$U(C)$ 0.20 (30)	$R(H,B,C,R)$ 0.68 (9)	$S(C,R)$ 0.09 (4)
1991	Red	0.14 (1)	0.83 (24)	$U(R)$ 0.39 (1)	$R(H,B,C,R)$ 0.55 (2)	$S(C,R)$ 0.21 (23)
1992	Total	0.04 (1)	0.94 (30)	$U(R)$ 0.24 (17)	$R(H,B,C,R)$ 0.71 (1)	$S(C,R)$ 0.16 (28)
1992	Blue	0.06 (1)	0.94 (30)	$U(R)$ 0.24 (21)	$R(H,B,C,R)$ 0.76 (2)	$S(B,R)$ 0.19 (28)
1992	Core	0.08 (1)	0.93 (30)	$U(R)$ 0.25 (30)	$R(H,B,C,R)$ 0.69 (2)	$S(H,C,R)$ 0.12 (30)
1992	Red	0.14 (1)	0.96 (30)	$U(R)$ 0.27 (1)	$R(H,B,C,R)$ 0.62 (2)	$S(H,C,R)$ 0.30 (30)
1993	Total	0.23 (1)	0.91 (23)	$U(H)$ 0.46 (1)	$R(H,C)$ 0.40 (5)	$S(H,B,R)$ 0.35 (22)
1993	Blue	0.28 (1)	0.91 (29)	$U(R)$ 0.42 (30)	$R(H,B,C)$ 0.41 (4)	$S(B,C,R)$ 0.44 (25)
1993	Core	0.20 (1)	0.83 (23)	$U(R)$ 0.63 (27)	$R(H,B,C)$ 0.24 (11)	$S(H,B,C,R)$ 0.30 (19)
1993	Red	0.23 (1)	0.81 (13)	$U(R)$ 0.75 (1)	$R(H,B,C,R)$ 0.34 (29)	$S(H,B,C,R)$ 0.25 (16)

Table 3. As Table 2 for the variable set F5100, blue, core and red Hβ. F denotes the continuum.

Epoch	Target	$\ell_{\rm min}$ (d)	$\ell_{\rm max}$ (d)	Largest U	Largest R	Largest S
1989	F5100	0.22 (1)	0.71 (18)	$U(F)$ 0.88 (4)	$R(B,C)$ 0.24 (30)	$S(F,B)$ 0.23 (15)
1989	Blue	0.12 (8)	0.45 (29)	$U(F)$ 0.75 (27)	$R(F,B,C,R)$ 0.35 (11)	$S(F,R)$ 0.26 (16)
1989	Core	0.05 (24)	0.40 (30)	$U(F)$ 0.87 (23)	$R(F,B,C,R)$ 0.40 (10)	$S(F,C)$ 0.18 (13)
1989	Red	0.17 (1)	0.50 (30)	$U(F)$ 0.83 (20)	$R(F,B,C,R)$ 0.39 (11)	$S(F,C)$ 0.24 (30)
1990	F5100	0.22 (1)	0.65 (26)	$U(F)$ 0.82 (1)	$R(B,C,R)$ 0.24 (30)	$S(F,B)$ 0.41 (20)
1990	Blue	0.06 (1)	0.63 (30)	$U(F)$ 0.80 (23)	$R(F,B,C,R)$ 0.42 (10)	$S(F,C)$ 0.21 (30)
1990	Core	0.08 (1)	0.71 (30)	$U(F)$ 0.75 (27)	$R(B,C,R)$ 0.44 (1)	$S(F,B)$ 0.25 (12)
1990	Red	0.09 (1)	0.73 (30)	$U(F)$ 0.62 (29)	$R(B,C,R)$ 0.44 (1)	$S(F,C)$ 0.25 (30)
1991	F5100	0.30 (1)	0.85 (18)	$U(F)$ 0.79 (1)	$R(B,R)$ 0.24 (30)	$S(F,B,C,R)$ 0.22 (20)
1991	Blue	0.03 (1)	0.45 (28)	$U(F)$ 0.29 (23)	$R(B,C,R)$ 0.56 (4)	$S(F,B)$ 0.24 (28)
1991	Core	0.00 (4)	0.46 (30)	$U(C)$ 0.22 (1)	$R(B,C,R)$ 0.46 (5)	$S(F,B)$ 0.21 (23)
1991	Red	0.12 (1)	0.64 (26)	$U(F)$ 0.44 (18)	$R(B,C,R)$ 0.41 (1)	$S(F,B,R)$ 0.18 (29)
1992	F5100	0.28 (1)	0.84 (30)	$U(F)$ 0.55 (6)	$R(F,B,C,R)$ 0.29 (1)	$S(F,R)$ 0.34 (30)
1992	Blue	0.06 (1)	0.75 (30)	$U(F)$ 0.62 (21)	$R(F,B,C,R)$ 0.52 (7)	$S(F,B)$ 0.21 (28)
1992	Core	0.07 (1)	0.75 (30)	$U(F)$ 0.41 (17)	$R(F,B,C,R)$ 0.54 (4)	$S(F,C)$ 0.21 (28)
1992	Red	0.11 (1)	0.81 (30)	$U(F)$ 0.61 (16)	$R(F,B,C,R)$ 0.54 (3)	$S(F,B)$ 0.24 (25)
1993	F5100	0.38 (1)	0.84 (30)	$U(F)$ 0.85 (1)	$R(C,R)$ 0.31 (27)	$S(F,B,C,R)$ 0.33 (28)
1993	Blue	0.28 (1)	0.70 (30)	$U(F)$ 0.61 (26)	$R(B,C)$ 0.38 (4)	$S(F,C)$ 0.25 (12)
1993	Core	0.19 (1)	0.61 (19)	$U(C)$ 0.70 (1)	$R(F,R)$ 0.34 (28)	$S(F,R)$ 0.40 (28)
1993	Red	0.24 (1)	0.77 (16)	$U(R)$ 0.87 (1)	$R(B,C,R)$ 0.18 (28)	$S(F,B,C,R)$ 0.36 (13)

Continuum to line information. The F5100 set isolates what the continuum adds to a line target beyond the line components: the unique term $U(F)$. For the blue, core and red targets it rises to a maximum of 0.29 to 0.87 (two bins) at lags of 16 to 29 days in 13 of 15 curves (Table 5). The two exceptions have no continuum term above 0.05: the core target in 1991 and the red target in 1993. With three bins the exceptions are the blue and core targets in 1991. A unique continuum component that peaks at a lag of tens of days and leaves leakage near its minimum (0.05 for the 1989 core target at 24 days) is what a delayed response to the continuum would produce. The same peaks give a velocity ordering of the response that we compare with the published seasons below.

Epoch comparison. Mean leakage over lags and targets (common lags, two bins) is 0.49, 0.53, 0.43, 0.58 and 0.65 for 1989 to 1993. These follow the ordering of the grid tuple counts (minimum 134, 161, 133, 211 and 231, respectively), with rank correlation one. Our categorical calibration shows that fewer tuples give a lower estimated leakage even for independent variables, so differences of leakage between epochs mainly reflect sample size.

3.3 Uncommon lags

Table 4 gives the mean leakage of the three lag configurations (two bins, all lags, all four targets of each set) and the change of the allocation. Figure 4 shows an example; the component figures for every configuration are in the atlas files.

Sampling the blue wing five days further back lowers the mean leakage relative to common lags in eight of the ten epoch and set combinations; the red wing does so in four (1989 and 1990), and raises it in 1991 to 1993. The mean differences are small, at most 0.08, but the allocation among components changes substantially: the median total variation $\frac{1}{2}\sum_a |f_a^{\mathrm{uncommon}} - f_a^{\mathrm{common}}|$ is 0.30 for both configurations (0.23 to 0.24 for the F5100 set and 0.36 to 0.40 for the total set), and ranges from 0.22 in 1993 to 0.46 in 1989. The lag assigned to each predictor therefore matters for which components receive the information even when the total information changes little. Blue leading gives the lower mean leakage for all three line targets in 1991 and 1992 (both sets) and in the F5100 set in 1990 and 1993; in 1989 it is higher for most targets. This test uses all targets, including the continuum and the target's own history, so it is not a velocity resolved delay measurement.

3.4 Comparison with Lu et al. and Xi et al.

Lu et al. (2022) compile the Hβ lags of the five historical campaigns in their Table 6. Table 5 lists them beside the lag at which the unique continuum component of the blue, core and red targets is largest.

The median peak lag over the detected line targets is 23, 27, 20.5, 17 and 23 days (two bins) against the published 19.7, 18.6, 15.9, 11.0 and 13.0 days. It exceeds the published lag in every season by a median of 6.0 days (5.3 days with three bins). The rank correlation between the two sets of lags is 0.67 for two bins and 0.36 for three (five points, not significant). Both show shorter lags in 1991 and 1992 than in 1989 and 1990, and a longer lag in 1993 than in 1992 (17 to 23 days for two bins, 19 to 20 days for three bins). A maximum of a unique information fraction on a one day grid is not the centroid of a cross-correlation function; the information term also depends on the history of the target and on the broad transfer function, and a longer apparent lag is expected when the response is spread over time. The comparison therefore supports a continuum driven response with a delay of the right order in the bright historical state, not an independent lag measurement.

Xi et al. (2025) measure much shorter Hβ lags in the modern low luminosity seasons (Table 6) and show that the velocity resolved lag profile changes between them: an M-shaped profile in 2015, a red-leads-blue pattern in 2019 and 2021 interpreted as inflow, and a blue-leads-red pattern in 2020 and 2023 interpreted as outflow. We read the same ordering from the peak lags of Table 5. With two bins the red wing peaks earlier than the blue wing in 1989 (20 against 27 d), 1991 (18 against 23 d) and 1992 (16 against 21 d), the core peaks earlier than the blue wing in 1993 (20 against 26 d), and the order is reversed in 1990 (blue 23, core 27, red 29 d). With three bins the red peak is again earlier in 1989, 1992 and 1993, and is tied with the blue peak in 1990. A red-leads-blue ordering is therefore found in four of the five historical seasons and survives the change of bins in 1989, 1992 and 1993, similar to the signature that Xi et al. (2025) identify in 2019 and 2021; the season 1990 resembles their blue-leads-red seasons but is not robust to the bin count. The differences between wings are five to seven days, a handful of grid steps, and have no uncertainty. They do not by themselves establish radial inflow: the Wanders and Peterson intervals are fixed wavelength bands and not equal flux bins, the three line series share spectra, and the uncommon lag test of the previous subsection does not single out one wing ordering. The velocity ordering is nevertheless the information theoretic counterpart of the quantity on which Xi et al. (2025) base their kinematic classification, and it varies between seasons in our data as in theirs.

The historical lags of 11 to 20 days (Lu et al. 2022) are longer than the 7 to 10 days of 2015 to 2021 and comparable with the 15.6 days of 2023. Xi et al. (2025) show that the broad line region radius follows the optical luminosity with a delay of 3.5 years: the radius fell with the luminosity after the decline that reached its minimum around 2005 and 2010, and approached its typical historical value only by 2023. The short lags of 2015 to 2021 therefore belong to a contracted line region that has not yet recovered, and the historical seasons correspond to the typical radius to which it returned. At the lag of the continuum peak, the leakage of the line targets remains between 0.15 and 0.71 (median 0.54, two bins), so about half of the entropy of a line target is not accounted for by the monitored observables. A driving continuum that differs from the observed 5100 Å flux, as in the disc wind shielding proposed for the 2014 'BLR holiday' (cited in the introduction of Xi et al. 2025), is one possible reason; the noise of the fluxes and the coarse two and three state description are others.

3.5 Sensitivity to the number of bins

Changing the histograms from two to three bins changes the allocation substantially (Table 7). The median total variation $D_{\mathrm{TV}} = \frac{1}{2}\sum_a |f_a^{(2)} - f_a^{(3)}|$ between the bin settings is between 0.27 and 0.58 in all epochs and configurations. For common lags the mean leakage is lower for three bins (0.34 and 0.46 for the F5100 and total sets against 0.46 and 0.61), consistent with the larger finite sample information of sparse histograms. The position of the main peaks can move by ten days or more (Table 5), so the quantities that depend on a single peak lag should be read with this resolution.

3.6 Limits of the interpretation

The component fractions describe how the available predictive information is distributed; they do not by themselves identify causal coupling or a physical delay. The independent categorical benchmark below shows that finite samples of independent variables produce substantial apparent information, particularly with three states, and

Table 4. Mean leakage over the 30 lags and four targets for common lags and for the blue leading and red leading uncommon lags (two bins). The next column is the mean difference blue leading minus red leading over the three line targets. The last column counts line targets for which blue leading has the lower mean leakage.

Epoch	Set	Common	Blue lead	Red lead	$\Delta\ell$	Targets
1989	Total	0.607	0.616	0.605	+0.012	0 of 3
1989	F5100	0.366	0.365	0.355	+0.010	1 of 3
1990	Total	0.613	0.594	0.605	−0.008	2 of 3
1990	F5100	0.448	0.418	0.432	−0.019	3 of 3
1991	Total	0.481	0.458	0.557	−0.098	3 of 3
1991	F5100	0.388	0.398	0.452	−0.058	3 of 3
1992	Total	0.651	0.625	0.695	−0.069	3 of 3
1992	F5100	0.505	0.467	0.515	−0.052	3 of 3
1993	Total	0.702	0.696	0.703	−0.012	1 of 3
1993	F5100	0.601	0.568	0.620	−0.049	3 of 3

Table 5. Published Hβ lag against continuum ($\tau_{H\beta}$, days; Lu et al. 2022, Table 6) and lag in days of the maximum of $U(F)$, with its normalized value in parentheses, for the blue, core and red targets of the F5100 set with common lags. A dash means $U(F) < 0.05$ at every lag.

		Two bins			Three bins		
Epoch	$\tau_{H\beta}$ (Lu)	Blue	Core	Red	Blue	Core	Red
1989	$19.7^{+1.5}_{-1.5}$	27 (0.75)	23 (0.87)	20 (0.83)	25 (0.49)	27 (0.53)	19 (0.31)
1990	$18.6^{+2.1}_{-2.3}$	23 (0.80)	27 (0.75)	29 (0.62)	19 (0.35)	30 (0.35)	19 (0.35)
1991	$15.9^{+2.9}_{-2.5}$	23 (0.29)	–	18 (0.44)	–	–	13 (0.23)
1992	$11.0^{+1.9}_{-2.0}$	21 (0.62)	17 (0.41)	16 (0.61)	19 (0.26)	30 (0.25)	11 (0.36)
1993	$13.0^{+1.6}_{-1.4}$	26 (0.61)	20 (0.48)	–	27 (0.32)	20 (0.40)	12 (0.15)

Table 6. Published Hβ lags of the modern seasons and the kinematic classification of their velocity resolved lag profiles (Xi et al. 2025, Table 3 and Section 5.2; seasons before 2023 from Lu et al. 2022, updated by Xi et al. 2025 with a common method; for 2020 Lu et al. 2022 found no credible profile because of the very low variability).

Season	$\tau_{H\beta}$ (days)	Velocity resolved lag profile
2015	$7.2^{+1.3}_{-0.4}$	wings respond before the core ('M' shape), disk-like
2018	$7.0^{+2.3}_{-3.4}$	no strong ordering
2019	$8.9^{+2.0}_{-1.1}$	red leads blue, inflow-like
2020	$10.0^{+3.3}_{-3.3}$	blue leads red, outflow-like
2021	$9.0^{+1.9}_{-2.5}$	red leads blue, inflow-like
2023	$15.6^{+2.6}_{-2.9}$	blue leads red, outflow-like

that leakage is biased low at small sample sizes. Observation dates, interpolation across the gaps, flux errors, and the shared spectra of the line series are not included in that benchmark. Lag choices (1–30 d, five day offsets) and the 15 day interpolation limit are fixed in advance and were not tuned. No significance level has been computed for the scans.

An earlier analysis on the native observation dates, without interpolation, retained only four of 120 curves under a common date requirement across all lags. The regular grid above replaces it so that every variable can be a target in every epoch.

Table 7. Median total variation between two and three bin allocations over lags and targets, per epoch, for the F5100 set (first three columns) and the total set (last three), with common, blue leading and red leading lags.

	F5100 set			Total set		
Epoch	Common	Blue lead	Red lead	Common	Blue lead	Red lead
1989	0.37	0.51	0.57	0.39	0.48	0.44
1990	0.32	0.55	0.50	0.37	0.46	0.46
1991	0.55	0.58	0.49	0.49	0.51	0.53
1992	0.27	0.39	0.45	0.37	0.40	0.38
1993	0.42	0.45	0.48	0.36	0.36	0.37

Table 8. Median joint information I in bits and normalized leakage ℓ for independent categorical variables (200 draws per setting). The exact values are $I = 0$ and $\ell = 1$. These are not significance thresholds for the AGN data.

Tuples	I, two states	I, three states	ℓ, two states	ℓ, three states
36	0.353	1.304	0.641	0.157
78	0.158	1.076	0.839	0.313
90	0.136	1.015	0.862	0.353
512	0.019	0.268	0.981	0.830

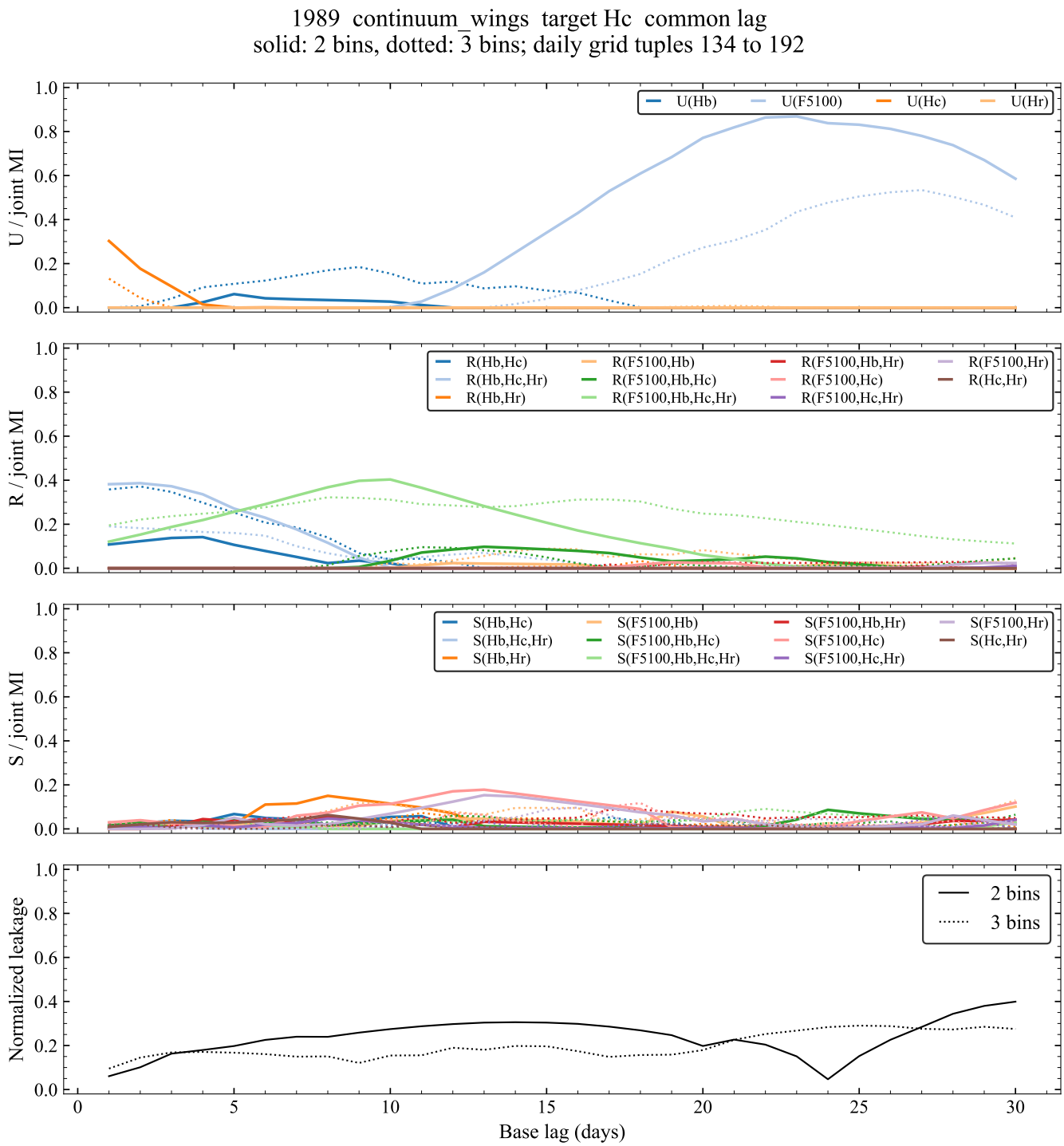


Figure 1. Campaign 1989, variable set F5100, blue, core and red Hβ, future target Core, common predictor lags. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 134 to 192 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

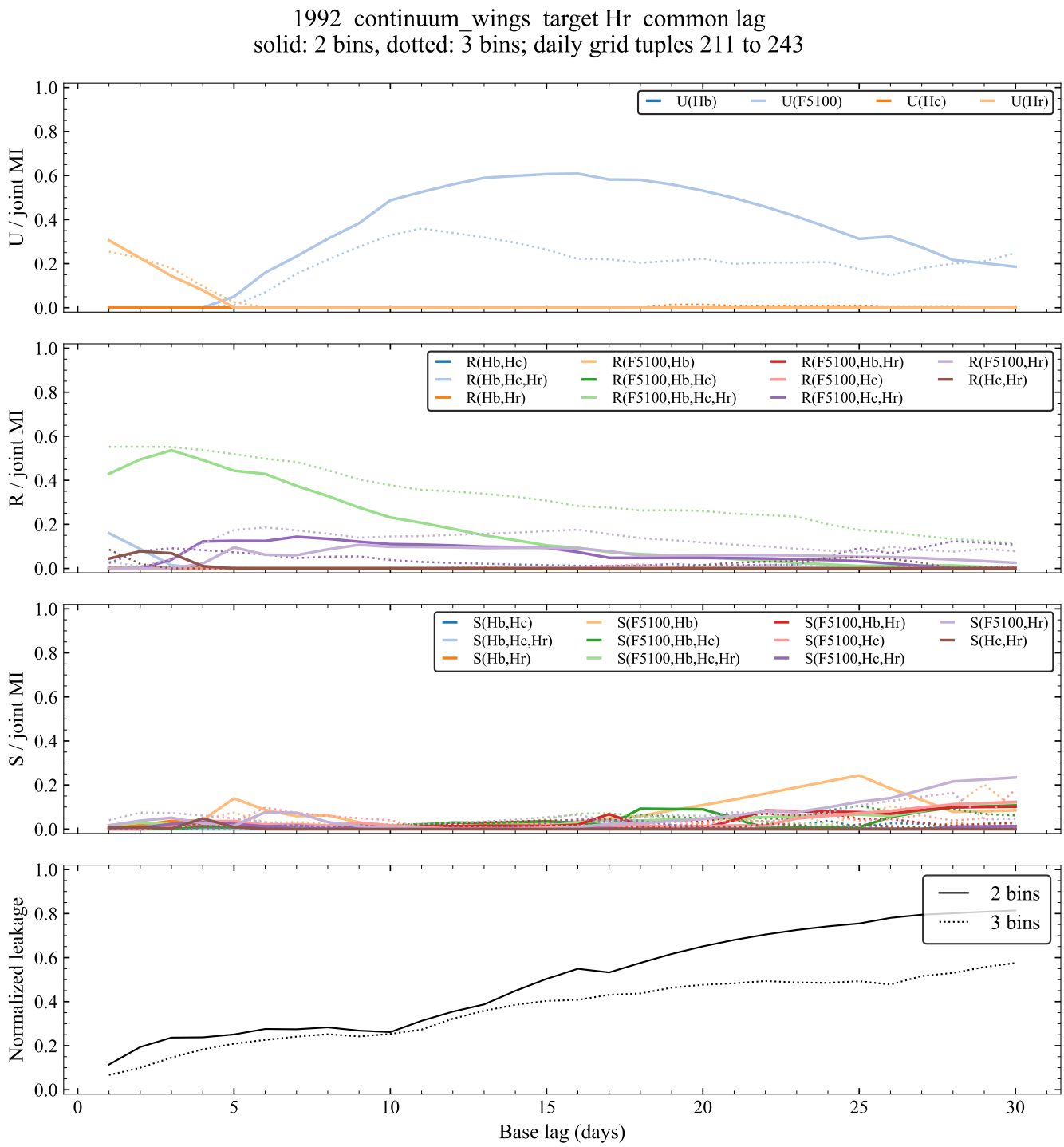


Figure 2. Campaign 1992, variable set F5100, blue, core and red Hβ, future target Red, common predictor lags. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 211 to 243 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

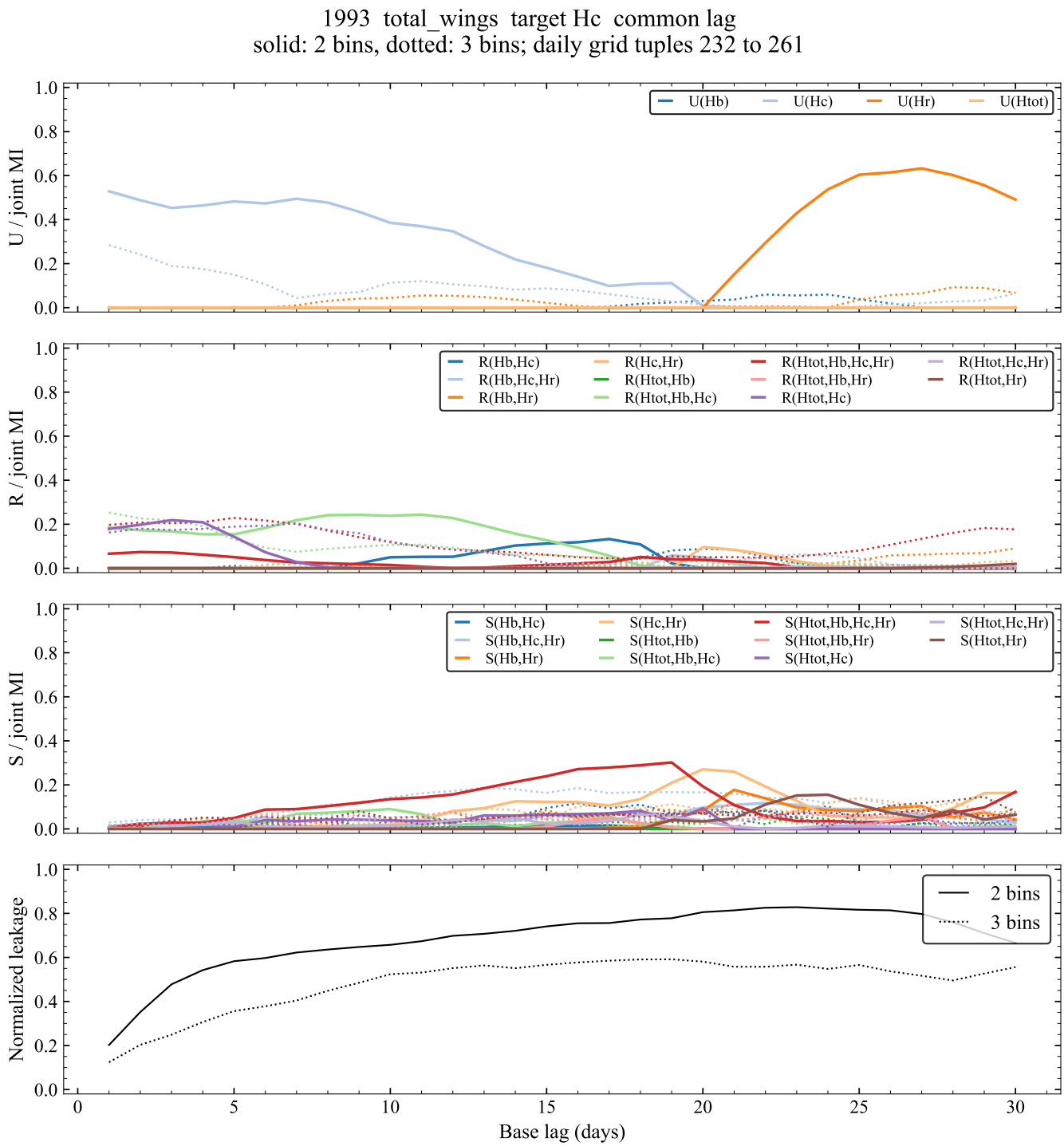


Figure 3. Campaign 1993, variable set total, blue, core and red Hβ, future target Core, common predictor lags. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 232 to 261 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

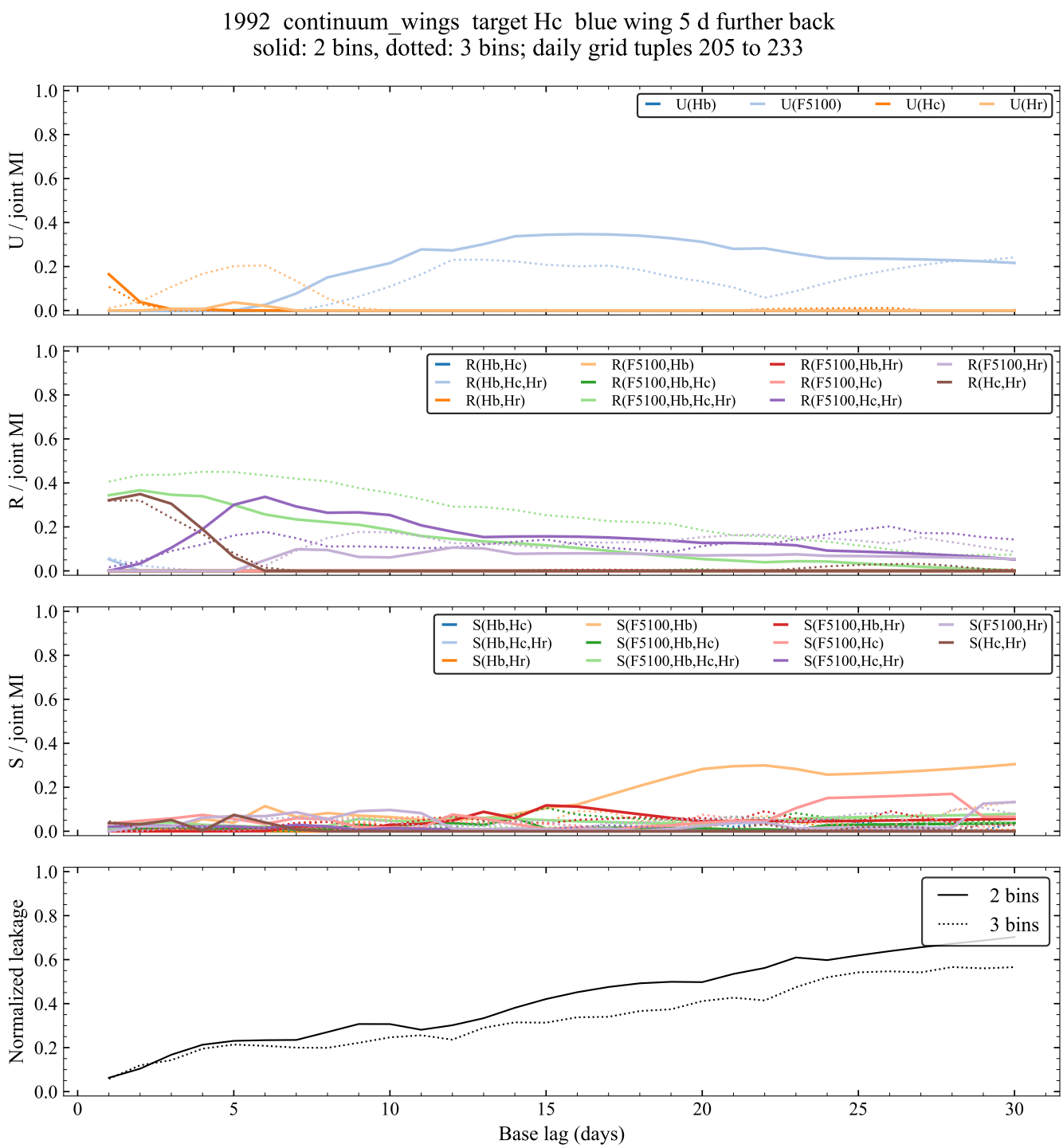


Figure 4. Campaign 1992, variable set F5100, blue, core and red Hβ, future target Core, uncommon predictor lags with the blue wing five days further back. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 205 to 233 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

4 DISCUSSION AND CONCLUSIONS

We decomposed the information that the 5100 Å continuum and the total, blue, core and red Hβ fluxes carry about the future of each of them, in each of the five historical seasons of NGC 5548, with 26 components and a leak at every lag from 1 to 30 d. The main results are the following.

(i) At short lags the target's own history carries most of the information, and the leak is lowest there (36 of 40 curves); at lags beyond 20 d the leak is highest (35 of 40) and approaches the level that independent categorical variables give in a finite sample (Table 8).

(ii) In the continuum set the unique continuum component of the blue, core and red targets peaks at 16–29 d in 13 of 15 curves. The peaks are longer than the published Hβ lags of the same seasons by a median of 6 d, and, like them, are shorter in 1991 and 1992 than in 1989 and 1990 and longer in 1993 than in 1992 (two bins).

(iii) The red wing peaks before the blue wing in 1989, 1991, 1992 and 1993, and after it in 1990. This is the red-leads-blue ordering associated with inflow in 2019 and 2021 (Xi et al. 2025), and its reversal in one season echoes the changing ordering reported by Lu et al. (2022) and Xi et al. (2025). The ordering of the 1990 season depends on the number of bins.

(iv) The allocation among components changes substantially with the number of histogram bins and with a five day shift of one wing. Mean leakage across seasons follows the number of grid samples, so it is not a comparison of the physical predictability of different seasons.

These are descriptive estimates. The fluxes are interpolated onto a daily grid, so the tuples are not independent; flux uncertainties are not propagated; no significance level is attached to any peak; and the three line series share spectra and the total is the sum of the other three. The wavelength bands are fixed and not equal flux bins. The wing ordering is therefore consistent with, but does not establish, radial inflow. A natural next step is the same analysis of the Lijiang seasons themselves, once the blue, core and red light curves are available, with a null calibration matched to the sampling.

DATA AND CODE AVAILABILITY

The NGC 5548 velocity-resolved spectra and optical continuum data are available from the AGN Watch compilation (http://www.astronomy.ohio-state.edu/~agnwatch/), and the corrected line fluxes are distributed with Wanders & Peterson (1996). The analysis code, the 26 components of every curve and the figures are available in the versioned repository at https://github.com/am610/AGN_SURD, folder collaborator_rebuild.

REFERENCES

Cover T. M., Thomas J. A., 2006, Elements of Information Theory, 2 edn. Wiley
Lu K.-X., et al., 2022, The Astrophysical Journal Supplement Series, 263, 10
Martínez Sánchez Á., 2025, Master's thesis, Massachusetts Institute of Technology
Martínez-Sánchez Á., Arranz G., Lozano-Durán A., 2024, Nature Communications, 15, 9296
Peterson B. M., et al., 2004, The Astrophysical Journal, 613, 682
Schreiber T., 2000, Physical Review Letters, 85, 461
Shannon C. E., 1948, Bell System Technical Journal, 27, 379
Wanders I., Peterson B. M., 1996, The Astrophysical Journal, 466, 174
Williams P. L., Beer R. D., 2010, arXiv preprint arXiv:1004.2515
Xi W.-Z., et al., 2025, The Astrophysical Journal, 995, 157